Car Rental Company

Initial Project Report

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Project Overview

The goal of the project is to design and implement a system for managing a car rental company. The application should support customer registration and profile editing, browsing available cars, making rental reservations, reviewing rental history, generating reports, online payments and using discount coupons. On the company side, the system should support fleet management, customer management, rental settlement, invoice generation and informing regular customers about promotions and discounts.

Selected Tech Stack

The project uses a Python-based stack. The selected technologies are presented in Table 1.

Layer	**Technology**	**Purpose**
Frontend	React	User interface for customers and rental company staff.
Backend API	Python + FastAPI	REST API, business logic, validation and documentation.
ORM	SQLAlchemy	Object-relational mapping between Python models and the database.
Migrations	Alembic	Versioned database schema changes.
Database	PostgreSQL	Persistent relational data storage.
Containers	Docker Compose	Local development environment and service orchestration.

Table 1: Selected project technology stack.

Initial Functional Scope

Actor	**Key functions**
Customer	Register and edit personal data, browse available vehicles, create reservations, review rental history, pay online and use discounts.
Staff	Manage cars and customers, settle rentals, issue invoices and communicate promotions to regular customers.

Table 2: Condensed functional scope.

Database Diagram

The current database design is centered around the `RENTAL` table, which connects customers, cars, pickup and return locations. Financial settlement is represented separately by the `INVOICE` table, while vehicle attributes, rental states and payment data are normalized through dedicated lookup entities.

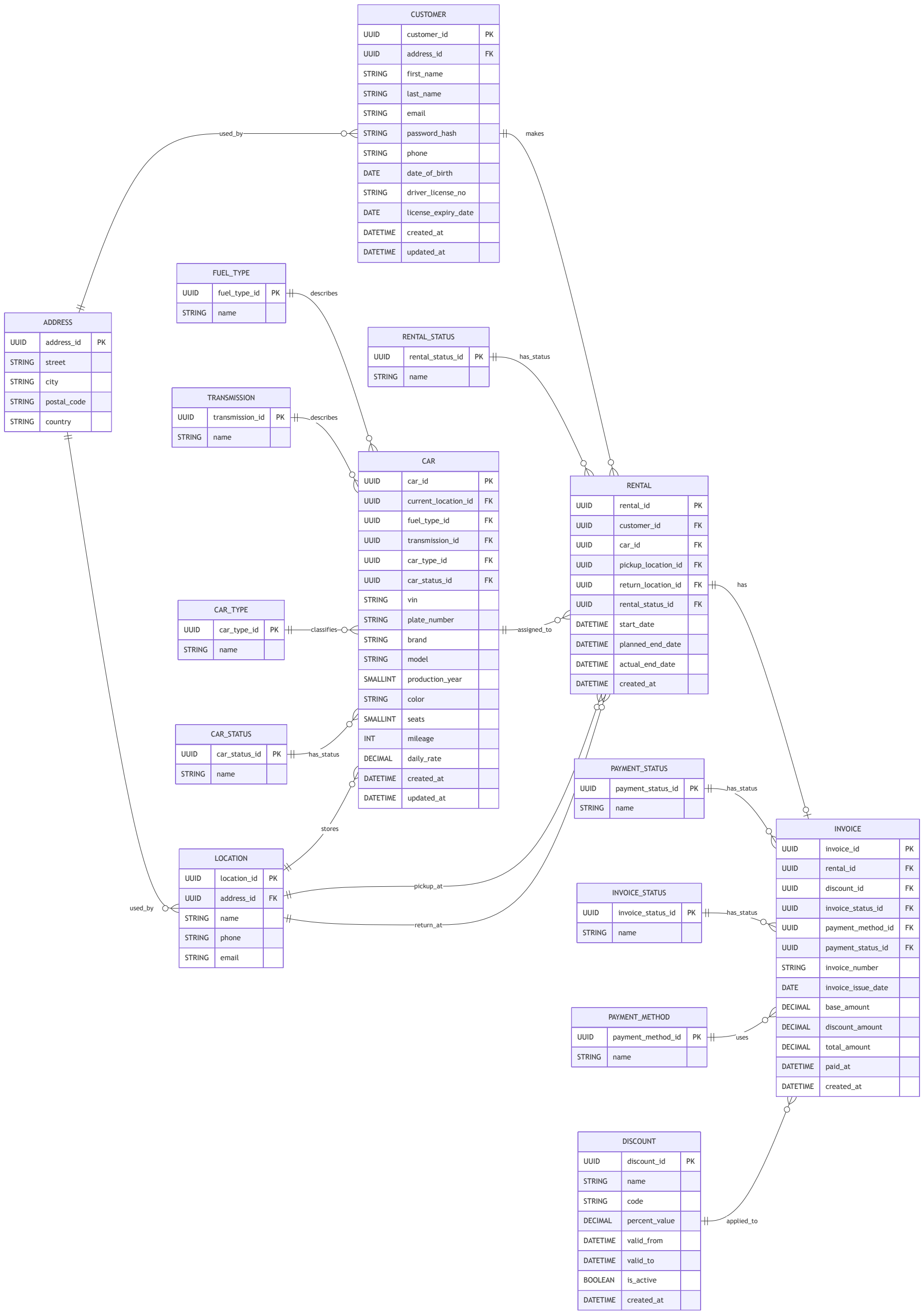

Figure 1: Database diagram in table form.

Core Entities

Entity	Role in the system
`CUSTOMER`	Stores personal, contact, authentication and driving license data required for rentals.
`ADDRESS`	Stores reusable address data used by customers and rental locations.
`LOCATION`	Represents rental locations with contact data and an address, used as car storage, pickup and return places.
`CAR`	Stores vehicle identity, descriptive attributes, mileage, daily rate, current location and references to vehicle lookup entities.
`FUEL_TYPE`	Lookup entity describing the fuel type assigned to a car.
`TRANSMISSION`	Lookup entity describing the transmission type assigned to a car.
`CAR_TYPE`	Lookup entity classifying cars by body or business category.
`CAR_STATUS`	Lookup entity describing the operational status of a car, for example available or unavailable.
`RENTAL`	Central transactional entity describing reservation and rental execution, including customer, car, pickup and return locations, status and rental dates.
`RENTAL_STATUS`	Lookup entity describing the state of a rental.
`INVOICE`	Stores settlement data for a rental, including invoice number, amounts before and after discount, issue date, payment method, payment status and payment timestamp.
`PAYMENT_STATUS`	Lookup entity describing whether an invoice payment is pending, completed, failed or otherwise classified.
`PAYMENT_METHOD`	Lookup entity describing the payment method used for an invoice.
`DISCOUNT`	Stores discount codes and percentage-based promotions with validity dates and active flag; applied to invoices when used.

Table 3: Main entities visible in the database diagram.

Conclusion

This first report presents the agreed project direction in a concise form: the project scope, the selected Python-based technology stack and the initial relational database design.